The sharp corner threshold for five-colour cubes

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Abstract

Every collection of fifteen cubes, each displaying all five colours of a fixed palette, contains eight that form a $2 \times 2 \times 2$ cube with monochromatic exterior faces using those five colours. Fourteen need not suffice: a counterexample uses just three varieties, with multiplicities $4, 5, 5$. This determines the five-colour corner value listed as unknown by Berkove, Eickemeyer and Schulman (2018). The lower bound has an elementary proof. For the upper bound, anchored Hall deficiencies and a shared-corner argument give 357 finite cases. Containment, budget bounds and a deletion lemma give 279 smaller problems, each with a separately checked interval certificate. Lean proves the interval and packed checkers sound, verifies all 279 certificates and the fourteen-cube obstruction, and connects the finite problems to physical cube arrangements, yielding a complete formal proof of the sharp threshold. A Hall argument then shows that any n^3 five-colour cubes can all form a complete n-cube for every $n \geq 3$, and gives the sharp frame value $12n - 16$ for $n \geq 10$. The sharp corner thresholds with at most one, two, or at least three varieties are respectively $8, 13, 15$. These extensions are also formalised.

AI transparency. The research, code and manuscript were developed using GPT-6-Astra (OpenAI), through Codex, under the author's direction. This working manuscript has not received independent human peer review; Section 8 states the formal verification scope.

1 Introduction

MacMahon's coloured-cube puzzles [1] lead to a family of extremal questions: how many face-coloured cubes guarantee that eight can be chosen to make a larger cube with monochromatic faces? Each small cube here uses all colours of the specified palette. Colours remain fixed, proper rotations identify varieties, and reflections are not allowed as moves. Collections may contain repeated varieties. Only exterior faces are constrained; touching interior faces need not have equal colours.

Berkove, Eickemeyer and Schulman [2, Section 8, Table 7] tabulate the corresponding frame numbers for palettes of two to six colours and leave the five-colour column unknown. Their corner values for two, three, four and six colours are respectively $11, 23, 11, 24$. We determine the missing five-colour value; these thresholds need not increase with the palette size. For a $2 \times 2 \times 2$ cube the frame consists of its eight corners. Let N_5 denote the least integer such that every collection of N_5 five-colour cubes contains a solution using all five colours on the target.

Theorem 1. *The sharp five-colour corner threshold is* $N_5 = 15$.

Theorem 10 bounds the frame numbers for every side length and determines them for $n \geq 10$, using an elementary Hall argument. Theorem 11 gives complete assemblies from any n^3 cubes for all $n \geq 3$. Theorem 13 determines the corner threshold when the number of source varieties is bounded, and shows that the fourteen-cube obstruction has minimum support. The corner conclusion for larger collections follows by choosing any fifteen cubes. The theorem concerns a target that may depend on the collection; it does not assert the ability to construct every target.

The proof has two distinct parts. A short corner argument excludes every target for an explicit fourteen-cube collection. For the upper bound, Hall's theorem gives linear alternatives for failure of a target. We cover hypothetical fifteen-cube counterexamples by 357 cases. Containment reduces these to 308, and initial budget bounds exclude another 29. A deletion lemma reduces the totals and budgets in the remaining 279 problems, which are excluded by finite interval certificates. The certificates, C++ programs, geometric audits, manuscript and Lean sources are included in the accompanying review archive.[1]

The companion paper [4] considers four-colour sources and targets using exactly three colours. It gives the sharp frame threshold $16n - 3$ for every side length $n \geq 2$, with a classification of the largest failing collections. Here both sources and targets use five colours, and the sharp corner result also gives the larger-cube conclusions above. The developments share the palette-independent geometry in `ColourCubes` [5]; neither threshold theorem is used to prove the other.

A limited literature search has found no later value for the five-colour entry. This is not a comprehensive priority claim. In particular, Rosenfeld's October 2026 article [6] has not been examined in full.

2 Geometry and Hall deficiencies

Represent a cube by its colours on the faces $(+x, -x, +y, -y, +z, -z)$ in a right-handed coordinate frame. A five-colour cube has one repeated colour. There are 75 varieties: for each repeated colour, three have its two faces opposite and twelve have them adjacent. In the opposite case, the other four colours form two opposite pairs, giving three possibilities. In the adjacent case, fix the two equal faces; the 24 orders of the other colours are identified in pairs by the rotation exchanging those equal faces. Thus there are 15 opposite-repeat varieties and 60 adjacent-repeat varieties. Each class is one orbit under colour permutations.

A corner type is the ordered triple of its incident face colours, modulo cyclic permutation. Order the outward face normals positively, equivalently read the triple counterclockwise from outside along the body diagonal. This reverses the clockwise convention in [2], consistently on every cube, and therefore preserves all matching and containment relations. Repeated occurrences of a corner type are retained as distinct target positions.

Lemma 2 (Corner–rotation correspondence)**.** *A source cube fits a target corner by a proper rotation if and only if it has the same cyclic corner type at some corner.*

Proof. A proper rotation preserves the orientation of a corner frame, so it induces a cyclic permutation of its three outward faces. Conversely, any two positively oriented corner frames, with a prescribed cyclic alignment, are related by a determinant-positive signed permutation of the coordinate axes. This is a proper rotation of the cube. □

The palette-independent Lean version of this lemma is reused from the shared `ColourCubes` library [5].

Fix a target T. Form a bipartite graph whose left vertices are its eight corner positions and whose right vertices are the individual source copies. A copy is adjacent to a position when it fits there. A matching covering the eight positions is exactly a solution: each chosen copy can be rotated independently, since interior faces are unrestricted. By Hall's theorem [3], failure is equivalent to a nonempty set D of target positions having fewer than $|D|$ neighbouring copies.

Index the varieties by $i = 1, \ldots, 75$, and let x_i be their multiplicities. For a set S of varieties, write $x(S) = \sum_{i \in S} x_i$. Let $N_T(D)$ be the varieties fitting at least one position of D. Then

[1] Intended repository: https://github.com/jasondavies/five-colour-cubes. Public release is pending; the review archive includes source checksums, the base Git revision and bundled pinned project dependencies.

failure of T is equivalent to the existence of a cut

$$x(N_T(D)) \le |D| - 1, \qquad \varnothing \ne D \subseteq \{1, \dots, 8\}. \tag{1}$$

There are 255 cuts per target, hence 19,125 cuts altogether.

3 An elementary fourteen-cube obstruction

Consider four copies of A, five of B, and five of C, where

$$A = (0, 0, 1, 2, 3, 4), \quad B = (0, 1, 2, 3, 2, 4), \quad C = (0, 3, 1, 2, 4, 1).$$

Their corner types are shown below. Triples denote cyclic classes; the last column counts positions whose type occurs in neither of the other varieties.

Variety	Corner types	Private positions
A	$013, 014, 023, 024, 031, 032, 041, 042$	5
B	$022, 023, 034, 042, 122, 124, 132, 143$	6
C	$011, 014, 021, 042, 113, 123, 134, 243$	6

Lemma 3. *The collection $4A + 5B + 5C$ has no five-colour solution.*

Proof. A solvable target must have every corner type in the displayed union. Its only two-colour types are $022, 122, 011, 113$. An adjacent-repeat target therefore repeats either colour 2, with types $022, 122$, or colour 1, with types $011, 113$. In either case these two corners determine the colours on the faces at the ends of the repeated-colour edge. The two orders of the remaining colours give a mirror pair: B and its mirror, or C and its mirror. The mirror of B requires 043, and the mirror of C requires 012; both types are absent. Only B, C remain.

For an opposite-repeat target, both orientations of each occurring unordered triple must be present. The unordered triples for which both cyclic orientations are available are

$$013, 014, 023, 024, 123, 134.$$

If its repeated colour is r, delete r from the four triples containing it. The remaining pairs must form a $K_{2,2}$ on the other four colours: choose one face from each of their two opposite pairs. Colours 2 and 4 occur in fewer than four listed triples. For $r = 1$ or $r = 3$, the four pairs have degree sequence $(3, 2, 2, 1)$, so do not form $K_{2,2}$. For $r = 0$, the pairs are $13, 14, 23, 24$, forcing opposite pairs $\{1, 2\}$ and $\{3, 4\}$. This is A.

Thus only A, B, C can be targets. Their private positions require respectively at least five, six and six copies of the corresponding variety, exceeding the available four, five and five copies (Figure 1). None is solvable. □

4 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 4 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\textit{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{2}$$

Conversely, such a set certifies failure of target A.

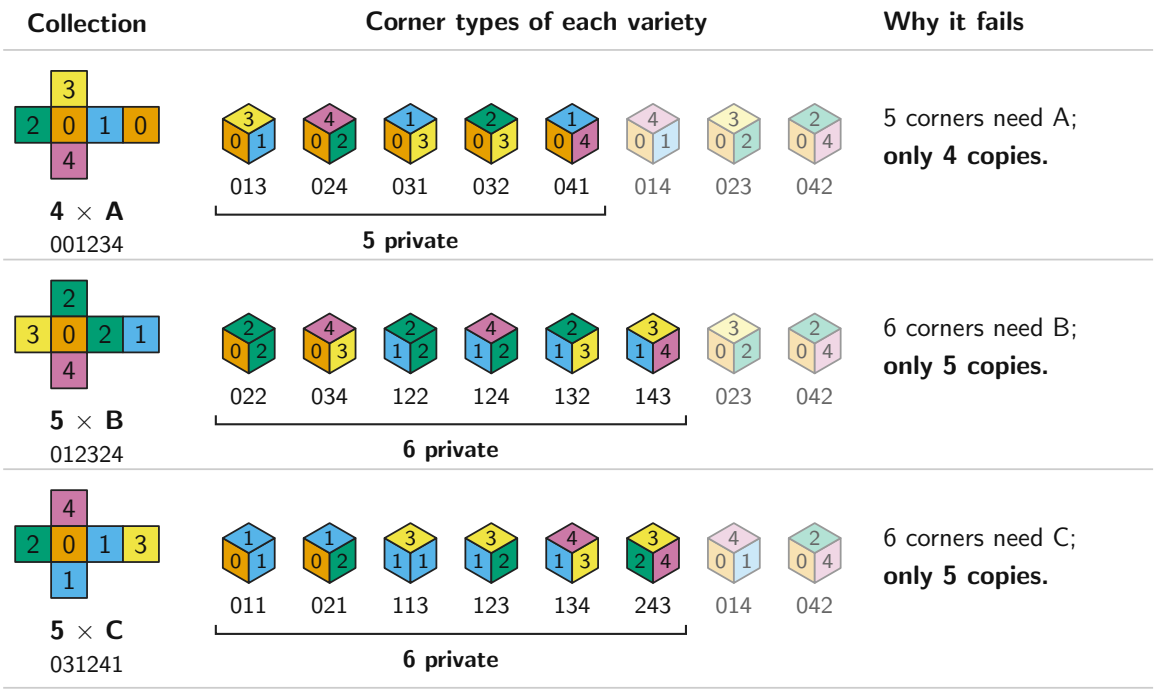


Figure 1: The fourteen-cube obstruction $4A + 5B + 5C$. Each net has $+x$ at its centre. Corner types are read counterclockwise from outside; the full-colour types occur on no other variety. A target of variety X has these private positions, and only copies of X fit them.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m - 1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 4, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If a copies of A and $8 - a$ copies of V build a target, with $a \le m$, then $x_V \le 7 - a$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least a, and V at least $8 - a$. They are also sufficient: at most a positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to a positions that A fits, and assign the remaining $8 - a$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The reported counts are constraint-orbit counts, not position-subset counts. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection is used only to transport a whole hypothetical counterexample, not as an allowed move on an individual cube. The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

For distinct collections a different choice of anchor reduces the count.

Lemma 5 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the anchor's two two-colour positions cannot be Hall-deficient.*

Proof. Each adjacent-repeat cube has exactly two such types, *aab* and *aac*, with $b \neq c$. There are only $5 \cdot 4 = 20$ types *aab*. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 5 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 6. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \tag{3}$$
$$\text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1.$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 7 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \le b$, and total 15, with the anchor outside S. For $0 \le s \le b$, put $G = 15 - b + s$ and suppose $G \ge m$. Every admissible vector x has a vector $y \le x$ with the same anchor multiplicity, total G, budget $y(S) \le s$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, s)$ on S.*

Proof. First retain only $\min(x(S), s)$ copies in S. This removes at most $b - s$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \ge m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \le s$ give $y_i \le s$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + s.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + s$ other copies; in particular, $G \ge m$ is automatic. Apply the lemma after selecting and normalising the original root, retaining its bounds. No new anchor selection is required for the smaller collection. Of the 279 remaining roots, 277 admit an exclusion certificate with $s = 0$. The other two, original cases 68 and 69, use $s = 1$ and $G = 12$. Their $s = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

5 Interval certificates

For a projected problem, admissibility is defined by (3) with the tightened upper bounds, total G and budget s. We certify that no admissible vector exists.

Write a box as $\ell \le x \le u$. For a cut $x(S) \le r$, two necessary conditions for feasibility in the box are

$$\ell(S) \le r, \qquad G \le r + u(S^c). \tag{4}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $s = r - \ell(S)$ and $g = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \le \min(u_i, \ell_i + s) \quad (i \in S), \qquad x_i \ge \max(\ell_i, u_i - g) \quad (i \notin S). \tag{5}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (4). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let s_*, g_* be the largest corresponding s, g. Then all vectors in the box that fail this target remain in the refined box

$$u'_i = \begin{cases} \min(u_i, \ell_i + s_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell'_i = \begin{cases} \max(\ell_i, u_i - g_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{6}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (5) imply (6). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell'_i = \max\left(\ell_i, G - \sum_{j \ne i} u_j\right), \qquad u'_i = \min\left(u_i, G - \sum_{j \ne i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (4). These are the checker's 77 possible closing reasons. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \le v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \le v$ or $x_i \ge v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 8 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

The projected archive contains 499,829 instructions in 313,094 compressed bytes, reducing the instruction count by 95.69% from the original archive. The repository documents both archives and the earlier search computations. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. The witness satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. The fixture is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

Proof of Theorem 1. Lemma 3 gives $N_5 \ge 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 6, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 7 supplies an admissible vector in its projected problem, which Proposition 8 and the sound interval deductions exclude. Hence $N_5 \le 15$. □

6 Frames and complete cubes

An n-frame consists of the corners and edges of an $n \times n \times n$ cube, and uses $12n - 16$ pieces. Let $F_5(n)$ be the least available pool size guaranteeing such a frame, and $G_5(n)$ the least pool size guaranteeing a complete n-cube, in each case with monochromatic exterior faces displaying all five colours. An assembly may use a subset of the pool.

Lemma 9 (Frame compatibility)**.** *Put $q = n - 2 \ge 1$. Every source variety fits at least $8q + 2$ positions of every five-colour n-frame target, and at least $9q + 2$ if the target is opposite-repeat. An adjacent-repeat source lacks exactly one bichromatic edge type. An opposite-repeat source lacks exactly two, which are disjoint; these two missing pairs determine its variety uniquely.*

Proof. A missing bichromatic pair must occupy opposite singleton faces: if a colour is repeated, a face of the other colour cannot be opposite both copies. An adjacent-repeat variety has one opposite pair of singleton faces, and an opposite-repeat variety has two. Two disjoint missing pairs specify two opposite face pairs, leaving the fifth colour on the remaining opposite faces, which determines a unique variety.

Write c, e for the numbers of target corners and geometric edges fitted by the source. Whenever either variety is opposite-repeat, $c \ge 2$. To see this, let that variety repeat r, with singleton opposite pairs $\{a, b\}$ and $\{c, d\}$. It contains both cyclic orientations of each rxy with $x \in \{a, b\}$ and $y \in \{c, d\}$. In the other variety, if an r-face has no adjacent r-face, its four surrounding faces contain at least three distinct colours. Their cyclic order crosses between the two pairs at least twice, giving two shared corner types. If the r-faces are adjacent, the other neighbours of each form a path of three distinct colours. Each path crosses between the pairs and gives a shared corner; these corners have different types.

A bichromatic target edge type occurs at most twice, and a doubled type contains the target's repeated colour. Two disjoint types therefore account for at most three target edges. There is at most one monochromatic target edge. These observations give the following bounds; A and O denote adjacent-repeat and opposite-repeat varieties.

Target	Source	c	e	$c+e$
O	O	≥ 2	≥ 9	≥ 11
O	A	≥ 2	≥ 10	≥ 12
A	O	≥ 2	≥ 8	≥ 10
A	A	≥ 0	≥ 9	≥ 10

Only the final entry needs an orientation argument. If an adjacent-repeat source fits exactly nine edges of an adjacent-repeat target, it misses the monochromatic edge and a doubled bichromatic type. Relabel colours and rotate to put the target at $(0, 1, 0, 2, 3, 4)$ and the missing type at 03. The source's 0- and 3-faces are opposite; its surrounding colours are $1, 2, 4$, one repeated on adjacent faces. Suppress this adjacent repetition. One of the two cyclic orders gives the shared target corner 123; the other gives 024 and 041. Thus $c \geq 1$ when $e = 9$.

Each geometric edge contributes q positions. Since $e \geq 8$ and $c + e \geq 10$,

$$c + qe = (c + e) + (q - 1)e \geq 8q + 2.$$

For an opposite-repeat target, $e \geq 9$ and $c + e \geq 11$ instead give $9q + 2$. □

Theorem 10. *For every integer $n \geq 3$,*

$$12n - 16 \leq F_5(n) \leq \max\{12n - 16,\ 11n - 6\}.$$

Consequently $F_5(n) = 12n - 16$ for $n \geq 10$.

Proof. The lower bound is the frame's piece count. Put

$$q = n - 2, \quad L = 12q + 8, \quad d = 4q + 6, \quad h = 3q + 6, \quad N = \max\{L, 11q + 16\}.$$

By Lemma 9, a source misses at most d target positions, or at most h for an opposite-repeat target. Consider N source copies. Call a bichromatic pair *rare* if at most $d - 1 = 4q + 5$ copies possess that edge type.

If two distinct pairs are rare, at least

$$N - 2(4q + 5) \geq 3q + 6 = h$$

copies lack both. The lemma implies that these pairs are disjoint and all these copies belong to one opposite-repeat variety V. Apply Hall's theorem to the L positions of target V and all N sources. Any nonempty position set of size at most h has these h copies as neighbours. Every set of size greater than h has every source as a neighbour, since no source misses more than h positions. As $N \geq L$, all Hall inequalities hold.

It remains to treat at most one rare pair. If there is none, reserve up to q cubes with an aa edge for each of the five colours a, taking all if fewer exist. These reserves are disjoint: a five-colour cube has only one repeated colour. At least $N - 5q \geq 6q + 16 \geq 15$ cubes remain, so Theorem 1 supplies a corner solution from them.

If exactly one pair ab is rare, first remove every source possessing it. At most $4q + 5$ copies are removed. Every remaining source has a, b on opposite singleton faces, so only the other three colours can be repeated. Reserve up to q monochromatic-edge cubes for each of those colours. At least

$$N - (4q + 5) - 3q \geq 4q + 11 \geq 15$$

cubes remain, and again choose a corner solution. Its target cannot have an ab edge, since either endpoint corner would require a source possessing that pair.

In both cases the chosen target uses no rare bichromatic type. If it has a monochromatic edge rr, its corner solution contains an rr-capable source outside the reserves. Thus the rr reserve reached size q; otherwise all such sources would have been removed. This reserve fills the monochromatic edge. Hence the corners and the possible monochromatic edge have a matching by distinct copies.

Return to all N sources and apply Hall to this target's full frame. A position set D containing no bichromatic edge position is covered by the preliminary matching. If D contains such a position and $|D| \le d$, the corresponding nonrare type has at least d neighbouring copies. If $|D| > d$, every source neighbours D by the missing-position bound. Again $N \ge L$ completes Hall's inequalities. The final matching may rearrange the preliminary corner placements.

Substituting $q = n-2$ gives the stated bound. For $n \ge 10$, its maximum is $12n-16$, proving equality. □

Theorem 11. *For every integer $n \ge 2$,*

$$G_5(n) = \max\{n^3, 15\}.$$

In particular, every collection of n^3 five-colour cubes can all be assembled into a complete n-cube for every $n \ge 3$.

Proof. For $n = 2$, a complete cube is its corner frame, so Theorem 1 gives $G_5(2) = 15$. For $n \ge 3$, the piece count requires $G_5(n) \ge n^3$. Conversely, $n^3 \ge 12n - 16$, since the difference counts face-interior and interior positions, and

$$n^3 - (11n - 6) = (n-3)(n^2 + 3n - 2) \ge 0.$$

Theorem 10 therefore supplies a frame from any n^3 sources. Each remaining face-interior position prescribes just one colour, which every unused cube has, and interior positions prescribe none. All remaining cubes can fill these positions. □

Thus any 27 cubes suffice for a $3 \times 3 \times 3$ assembly without spares. This does not determine the sharp frame value: it gives $20 \le F_5(3) \le 27$. The sharp frame values for $3 \le n \le 9$ remain undetermined here.

7 Collections with few varieties

Let $M(k)$ be the least number of cubes guaranteeing a corner solution when the collection contains at most k varieties.

Lemma 12. *Two five-colour varieties with disjoint corner sets can build an opposite-repeat target from four copies of each.*

Proof. This is a small finite geometric check. Of the $\binom{75}{2}$ unordered pairs, 330 have disjoint corner sets. Up to colour permutation and exchange of the sources, they have the following five representatives. The first source is 010234 throughout; tuples use the face order of Section 2. Each listed target has four corners supplied by each source.

Second source	Target
011243	001234
012324	012234
012334	012234
012434	012234
030412	001234

The accompanying geometric checker regenerates all varieties and verifies the orbit coverage and each witness. Lean checks a suitable target directly for every pair, without using the orbit reduction. □

Theorem 13. *For every integer $k \geq 1$,*

$$M(k) = \begin{cases} 8 & k = 1, \\ 13 & k = 2, \\ 15 & k \geq 3. \end{cases}$$

Consequently, three is the minimum number of varieties in a fourteen-cube collection without a solution.

Proof. Eight identical copies build their own variety, and fewer than eight pieces cannot form a corner solution. For two varieties, consider thirteen copies. If either variety occurs at least eight times, it builds itself. Otherwise the multiplicities are seven and six. If the varieties share a corner type, use a minority copy at that corner of the majority target and seven majority copies elsewhere. If their corner sets are disjoint, apply Lemma 12.

For the lower bound take six copies each of

$$P = (0, 1, 0, 2, 3, 4), \qquad Q = (0, 1, 1, 3, 2, 4).$$

Their corner sets share just 041. Inspection of the 75 varieties shows that only P, Q have all corner types in their union. Each has seven private positions, so neither can be built with six copies. The geometric checker verifies these assertions; Lean certifies failure using deficient corner sets for all 75 targets. Thus $M(2) = 13$. For $k \geq 3$, Theorem 1 supplies the upper bound, and the three-variety collection in Lemma 3 supplies the lower bound. □

Remark (A family of fourteen-cube obstructions). Keep A, C from Section 3, and put

$$B_1 = (0, 1, 2, 3, 2, 4), \quad B_2 = (0, 1, 2, 3, 3, 4), \quad B_3 = (0, 1, 2, 4, 3, 4).$$

All 21 collections

$$4A + 5C + x_1B_1 + x_2B_2 + x_3B_3, \qquad x_1 + x_2 + x_3 = 5, \quad x_i \in \mathbb{Z}_{\geq 0},$$

are unsolvable. The B_i share the six trichromatic corner types $023, 034, 042, 124, 132, 143$. The combined corner union allows only A, C, B_1, B_2, B_3 as targets, as a check of the catalogue confirms. Target A needs five A-copies, C needs six C-copies, and any B_i needs six copies from the B-group. The respective supplies are four, five and five. Lean verifies these shortages simultaneously for every distribution. The original example is $x_1 = 5, x_2 = x_3 = 0$.

8 Formalisation and reproducibility

The package uses Lean 4 and mathlib [7, 8], both at version `v4.35.0`. Lean proves soundness of the reference interval checker, of a faster arithmetic implementation, and of composition of bounded proof chunks. The faster implementation encodes a multiplicity vector as

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad B = 1024.$$

For a neighbourhood S, the weight $W(S) = \sum_{i \in S} B^{74-i}$ satisfies

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Lean proves this identity under $0 \le x_i \le 7$, including the absence of carries; the coefficient bound is $75 \cdot 7 < 1024$. It also proves decoding, coordinate updates and equivalence of the packed feasibility test to the integer inequalities. All 19,125 supplied weights are checked against their neighbourhood masks by the Lean kernel.

Chunk composition requires exact agreement of both boundary vectors and a proof excluding each referenced box; the exporter is untrusted. A limit of 128 instructions gives 4,786 chunks for the projected archive. All 279 projected certificates have passed kernel replay, together with their final assembly into the physical threshold and sharpness theorems. The exact case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The 100 declarations listed in `Audit.lean` and the final replay and extension theorems have audited dependencies confined to `propext`, `Classical.choice` and `Quot.sound`; no `native_decide` is used.

Lean proves completeness and uniqueness of the 75 cube varieties and checks all 45,000 compatibility bits using the shared library's palette-independent corner–rotation theorem. The physical lower-bound theorem, `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`, uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. In the formal proof, this finite check replaces the elementary target classification in Lemma 3; catalogue completeness and rotation transport extend it to every physical target.

Lean proves the physical Hall reduction, symmetry transport, anchor normalisation, two-variety bounds, anchor selection and the shared-corner argument. Finite checks certify the root capacities and cover representatives. Containment and initial budget tests reduce the 357 roots to 279. Lemma 7 is formalised by `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`; all concrete projection hypotheses and coverage of the retained roots are kernel-checked.

The projected archive's integer semantics are established in `FiveColour/ProjectedBridge.lean`. The identifications include all coordinate bounds, prescribed totals, budgets and Hall-cut families. This checks the entire admissibility predicate, not just the packed boundary vectors. The generated module `FiveColour/ProjectedChunks/Threshold.lean` assembles all 279 exclusions into the unconditional theorems `FiveColour.Packed.ProjectedChunks.threshold15` and `FiveColour.Packed.ProjectedChunks.sharpThreshold15`.

The frame extension uses general matching lemmas. These are formalised in `FiveColour/FiniteHall.lean`, and applied in `FiveColour/RareFrames.lean`. Lean checks the geometric compatibility bounds for all 5,625 source–target pairs and the unique-variety conclusion for two missing bichromatic pairs. The physical edge table is checked directly against face adjacency; corner counts reuse the verified incidence table. The Hall proof then applies to arbitrary $q = n-2 \ge 1$. Completion of a frame is proved in `FiveColour/FullCube.lean`, adapting the general construction from [4] without using its threshold theorem. `FiveColour/SmallSupport.lean` proves the two-variety bound and its sharpness, and `FiveColour/MixedObstructions.lean` proves the obstruction family by common Hall deficiencies. These additions need no new interval certificates.

The unconditional conclusions in `Solution.lean`, in namespace `FiveColour.Puzzle`, include `frame_bound`, `frame_threshold`, `full_cube`, `complete_cube_threshold` and `support_threshold`. Their axiom checks, including the 27-cube and 64-cube instances, are listed in `ExtensionsAudit.lean`.

The original and intermediate interfaces are documented in `FiveColour/README.md`. Reproduce the full proof and its axiom audit with:

```
make check WORKERS=2
make projected-replay-data WORKERS=8
make projected-replay WORKERS=8
make check-extensions
```

The review archive supplies certificates, code and reproduction commands. A Mathlib-only comparator challenge states the puzzle geometrically. Model equivalence is proved by

`FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution module applies the completed threshold theorem and builds in Lean. A successful full external comparator check has not yet been confirmed. Mathematical questions left open here include classifying all fourteen-cube obstructions and determining the sharp five-colour frame numbers for $3 \leq n \leq 9$. In particular, do twenty cubes always suffice for a 3-frame? The complete-cube numbers are determined by Theorem 11.

AI disclosure and review status

GPT-6-Astra (OpenAI), used through Codex, developed the arguments, search and verification programs, Lean formalisation and manuscript under the human author's direction. The author initiated and directed the research. Author review and submission approval of this working manuscript are pending. The inherited geometry has its own provenance in the pinned dependency. Automated checks and AI review are not independent human peer review.

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